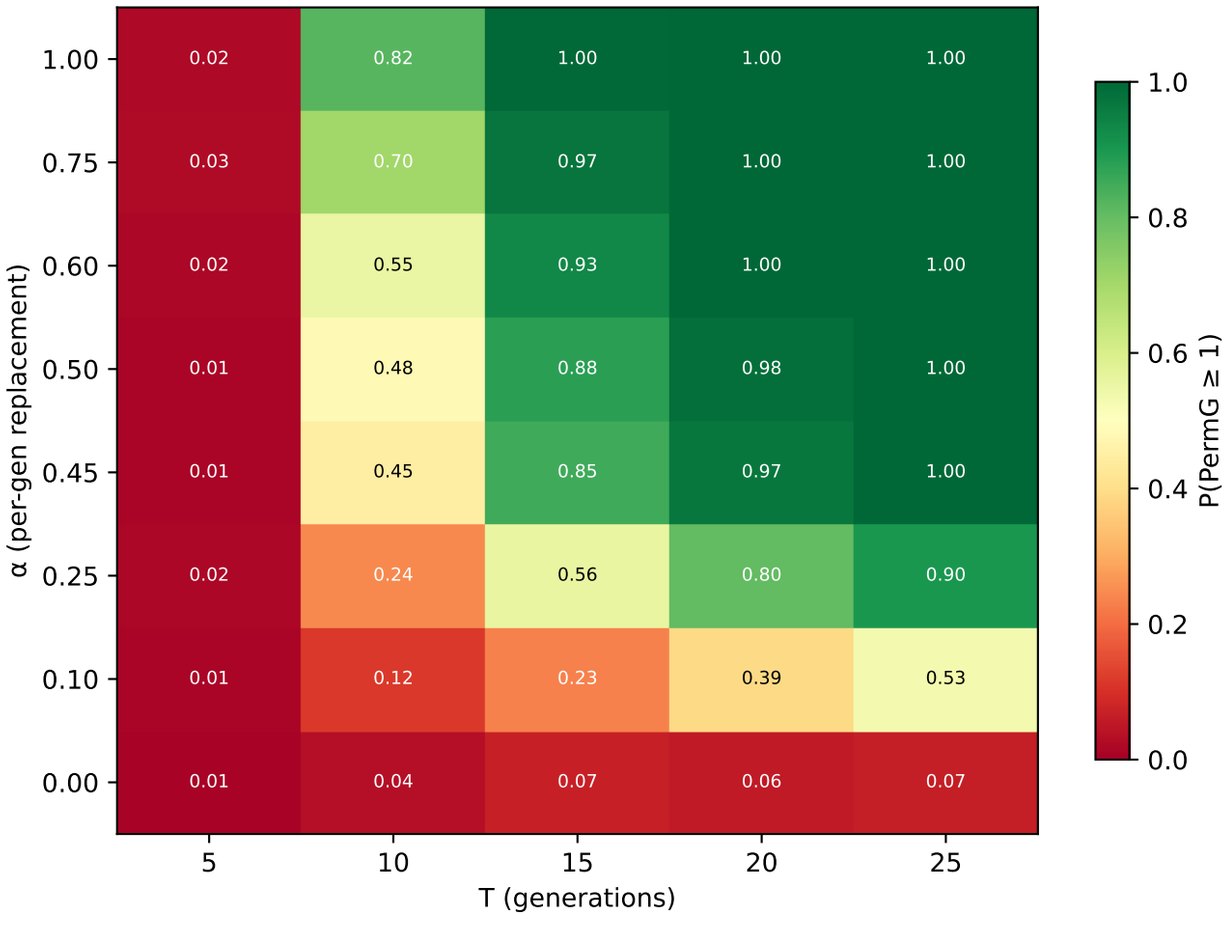
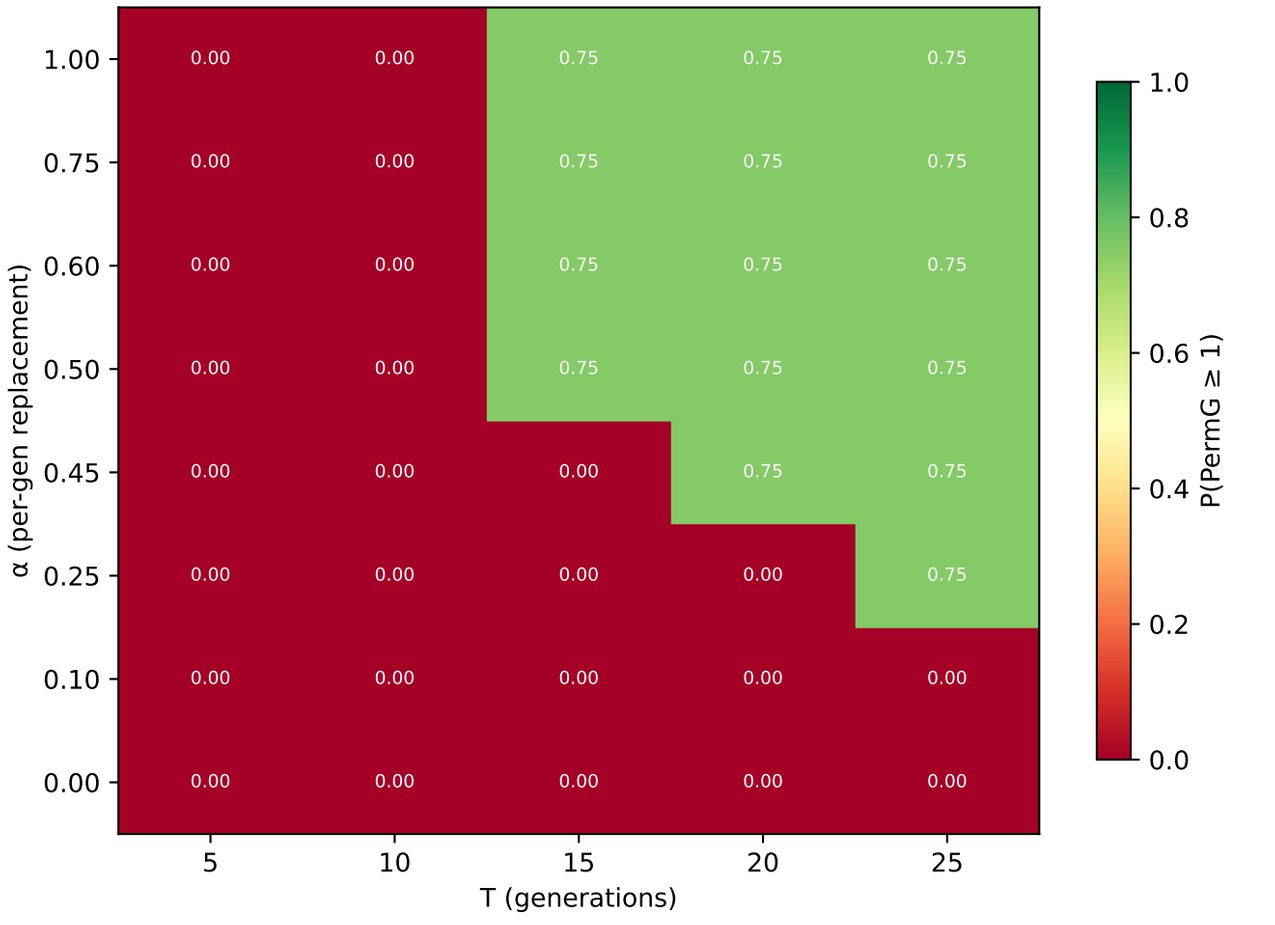
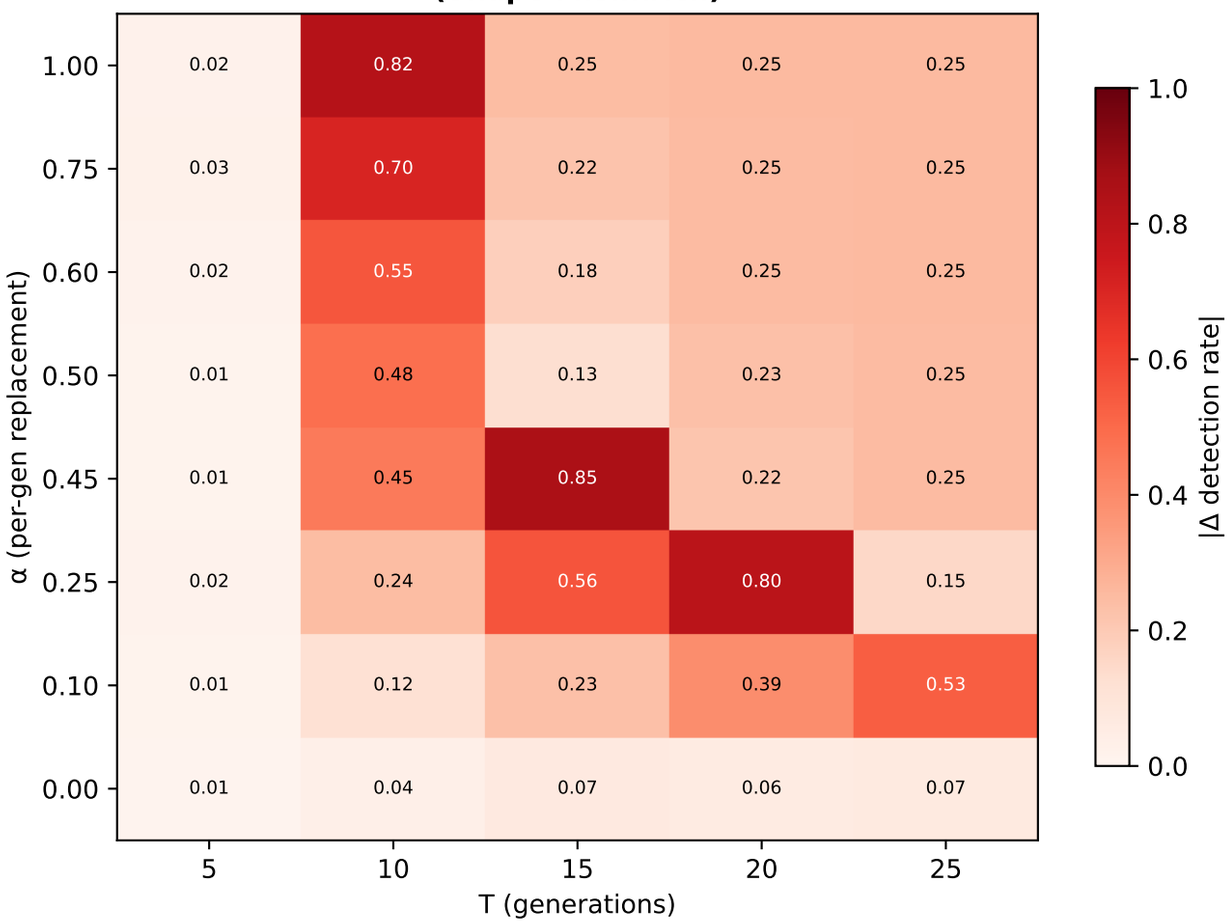
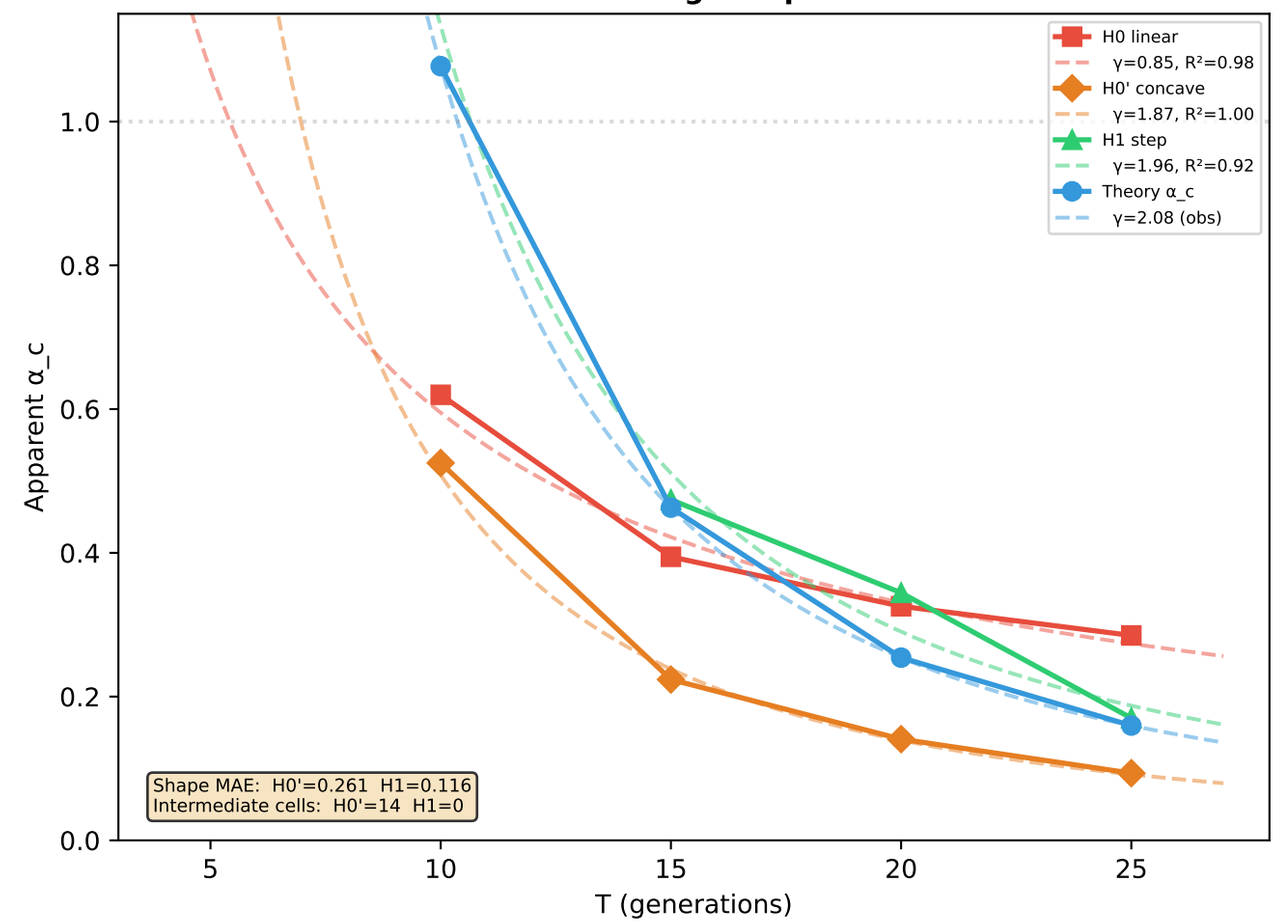

H0' Concave Signal (k=0.5)
Smooth Model Prediction
α (per-gen replacement)
T (generations)
P(PermG ≥ 1)
1.00 0.02 0.82 1.00 1.00 1.00
0.75 0.03 0.70 0.97 1.00 1.00
0.60 0.02 0.55 0.93 1.00 1.00
0.50 0.01 0.48 0.88 0.98 1.00
0.45 0.01 0.45 0.85 0.97 1.00
0.25 0.02 0.24 0.56 0.80 0.90
0.10 0.01 0.12 0.23 0.39 0.53
0.00 0.01 0.04 0.07 0.06 0.07
5 10 15 20 25
0.0 0.2 0.4 0.6 0.8 1.0


Observed Binary Pattern
(0/8 or 6/8, P_max=0.75)
α (per-gen replacement)
T (generations)
P(PermG ≥ 1)
1.00 0.00 0.00 0.75 0.75 0.75
0.75 0.00 0.00 0.75 0.75 0.75
0.60 0.00 0.00 0.75 0.75 0.75
0.50 0.00 0.00 0.75 0.75 0.75
0.45 0.00 0.00 0.00 0.75 0.75
0.25 0.00 0.00 0.00 0.00 0.75
0.10 0.00 0.00 0.00 0.00 0.00
0.00 0.00 0.00 0.00 0.00 0.00
5 10 15 20 25
0.0 0.2 0.4 0.6 0.8 1.0


|H0' − Observed| Residual
(Shape Mismatch)
α (per-gen replacement)
T (generations)
|Δ detection rate|
1.00 0.02 0.82 0.25 0.25 0.25
0.75 0.03 0.70 0.22 0.25 0.25
0.60 0.02 0.55 0.18 0.25 0.25
0.50 0.01 0.48 0.13 0.23 0.25
0.45 0.01 0.45 0.85 0.22 0.25
0.25 0.02 0.24 0.56 0.80 0.15
0.10 0.01 0.12 0.23 0.39 0.53
0.00 0.01 0.04 0.07 0.06 0.07
5 10 15 20 25
0.0 0.2 0.4 0.6 0.8 1.0


Onset Scaling Comparison
Apparent α_c
T (generations)
H0 linear
γ=0.85, R²=0.98
H0' concave
γ=1.87, R²=1.00
H1 step
γ=1.96, R²=0.92
Theory α_c
γ=2.08 (obs)
Shape MAE: H0'=0.261 H1=0.116
Intermediate cells: H0'=14 H1=0
0.0 0.2 0.4 0.6 0.8 1.0
5 10 15 20 25